
Quantum Computation and Communications (EE798V):

Problem Set 2

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Q. 1: For any two linear operators A, B on a Hilbert space, $\mathcal{H}$, the commutator $[A, B]$, and the anti-commutator $\{A, B\}$ operators are defined as follows.

$$[A, B] = AB - BA, \text{ and } \{A, B\} = AB + BA$$

Show the following results concerning the Pauli matrices $\sigma_x, \sigma_y, \sigma_z$.

- (a) $\{\sigma_x, \sigma_y\} = \{\sigma_y, \sigma_z\} = \{\sigma_z, \sigma_x\} = 0$.
- (b) $[\sigma_x, \sigma_y] = 2i\sigma_z, [\sigma_y, \sigma_z] = 2i\sigma_x, [\sigma_z, \sigma_x] = 2i\sigma_y$.

Therefore, the Pauli matrices are not commutative but anti-commutative.

Q. 2: Consider a point on the Bloch sphere that is characterized by the angles $(\phi, \theta) = (\pi/4, \pi/4)$.

- (a) Write the Bloch coordinates of the point.
- (b) Express the point as an element of the Hilbert space of dimension 2.
- (c) Assume that the state obtained in (b) passes through a σ_x gate. Write the resulting state as an element of the Hilbert space of dimension 2.
- (d) Compute the Bloch sphere angles (ϕ, θ) of the state computed in (d).
- (e) Repeat (c) and (d) assuming the state computed in (b) passes through a Hadamard gate.

Q. 3: Consider a 2-qubit composite system whose joint state is expressed as $|\psi\rangle = \frac{1}{\sqrt{2}}(1, 0, 1, 0)^\top$.

- (a) Show that the composite system is not entangled.
- (b) Assume that the qubits pass through unitary transformations σ_y and H respectively. If these transformations are not correlated, then compute the matrix representation of the transformation describing the evolution of the composite state.
- (c) Verify that the above transformation is unitary.
- (d) Show that if U, V are arbitrary unitary matrices, then $U \otimes V$ is also unitary.
- (e) Prove that the composite system is entangled if the joint state is described as $|\psi\rangle = \frac{1}{\sqrt{2}}(1, 0, 0, 1)^\top$.

Q. 4: Consider a 3-qubit composite system whose joint state is described as

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|000\rangle + |101\rangle + |110\rangle)$$

- (a) Assume that a partial measurement is done on the first (leftmost) qubit with respect to the orthonormal basis (ONB) $\{|0\rangle, |1\rangle\}$. Find out the probabilities of different outcomes, and the joint states after measurement.
- (b) Assume that the first (leftmost) and the third (rightmost) qubits are jointly measured, and the second qubit is left untouched. Let the ONB for measuring the first qubit is the same as that mentioned in (a), and that for measuring the third qubit is $\{|+\rangle, |-\rangle\}$ where

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \text{ and } |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Find out the probabilities of different outcomes and the corresponding joint states after measurements.

Q. 5: Consider a quantum system with m possible outcomes. Assume that a measurement is described by an ONB $\{|\psi_i\rangle\}_{i=1}^m$. Define $M_i = |\psi_i\rangle\langle\psi_i|$.

(a) Prove that any state $|\psi\rangle$ can be written as

$$|\psi\rangle = \sum_{i=1}^m M_i |\psi\rangle$$

(b) Argue that $\sum_{i=1}^m M_i$ is an identity transformation.

(c) If the above measurement is performed on the state $|\psi\rangle$, then show that the probability that outcome i occurs is given by $P_i = \langle\psi|M_i|\psi\rangle$.

(d) Show that $\{P_i\}_{i=1}^m$ is a valid probability distribution i.e., $P_i \geq 0$, $i \in \{1, \dots, M\}$, and $\sum_{i=1}^m P_i = 1$.

(e) Show that if the above measurement is performed on the state $\exp(i\phi)|\psi\rangle$ where $\phi \in [0, 2\pi]$, then the outcome probabilities remain the same. Hence, $|\psi\rangle$ and $\exp(i\phi)|\psi\rangle$ describe the same physical state.

(f) Show that if the above measurement is performed on the state $|\psi\rangle$, then the state after the measurement is given as $M_i|\psi\rangle/\sqrt{\langle\psi|M_i|\psi\rangle}$ if the corresponding outcome is i .